

Derek Corniello

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LinkedIn: linkedin.com/in/derek-corniello | GitHub: github.com/DerekCorniello

EDUCATION

University of Cincinnati, Cincinnati, Ohio

Expected Graduation: May 2027

Bachelor of Science, Computer Science

GPA: 3.9

Master of Engineering, Software Engineering

GPA: 4.0

Relevant Courses: Data Structures and Algorithms, Compiler Theory, Software Engineering, OS and Systems Programming

SKILLS

Languages: Python, Go, Rust, Java, TypeScript, JavaScript, C++, SQL

Development Tools: git, neovim, tmux, Linux, AWS, Unity

EXPERIENCE

Fifth Third Bank | Software Engineer Intern

January 2026 – May 2026

- Contributed to the modernization of an IVR system, migrating from a telecommunications platform to AWS infrastructure
- Developed 20+ backend API microservice components to support over 100,000 daily callers and customer data flow
- Contributed 30+ pull requests across full-stack services supporting system modernization efforts
- Participated in code reviews and cross-team collaboration, improving code quality and stability in a high-traffic system

Siemens Digital Industries Software | Software Engineering Intern

January 2024 – December 2025

- Collaborated with 3 cross-functional teams to design, develop, and test over 10 innovative software solutions
- Built REST APIs with sub-millisecond responses, operating on 100,000+ database entries efficiently on dozens of endpoints
- Implemented AI features using RAG, embeddings, graphs and ETL pipelines, enhancing LLM efficiency and accuracy
- Administered over 200 pages of documentation of projects, facilitating seamless knowledge transfer and collaboration

PROJECTS

Mux Programming Language | mux-lang.dev | Rust, LLVM

- A strong, statically-typed, compiled programming language designed to help programmers write simple, clean, and easy code
- Built core compiler pipeline including lexing, parsing, semantic analysis, and intermediate representation generation
- Created a runtime and standard library foundation to support core language functionality and developer usability

Quick Mouse | [Github](https://github.com) (First Place @ MakeUC 2025) | Go, React, Websockets

- A lightweight CLI application that connects your phone to your computer via a QR code to utilize your phone as a mouse
- Designed a cross-platform input transition layer converting mobile input gestures and movements into native mouse events
- Architected a device pairing system enabling QR-based authentication communicating over websockets

8 Bit Beats | [8BitBeats - crates.io](https://8BitBeats-crates.io) (1,000+ Downloads) | Rust, Terminal, Audio DSP

- A terminal-based chiptune generator with deterministic audio synthesis and real-time playback control
- Authored a multi-threaded audio generation system, using algorithms to create unique tracks in about 200 milliseconds
- Built a dynamic TUI with real-time controls, enabling users to create, preview, and replay 8-bit songs from seed IDs

CERTIFICATIONS AND HONORS

Amazon Web Services (AWS) - Certified Cloud Practitioner

February 2026

Institute of Electrical and Electronics Engineers: Eta Kappa Nu Honors Society – Executive Officer

October 2025 – May 2026

Dean's List

August 2023 – Present

Cincinnatus Scholar

August 2023 – Present